

Unit-2 Averages and Progressions

Study Guide

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2.1 Average

$$\text{Average} = \frac{\text{Sum of quantities}}{\text{Number of quantities}}$$

Example-1: A man purchased 5 toys at Rs.220 each, 6 toys at Rs. 250 each and 9 toys at Rs. 300 each. Calculate the average cost of 1 toy.

Solution: Price of 5 toys = $200 \times 5 = 1000$

Price of 6 toys = $250 \times 6 = 1500$

Price of 9 toys = $300 \times 9 = 2700$

Total number of toys = $5 + 6 + 9 = 20$

$$\text{Average Price of 1 toy} = \frac{1000+1500+2700}{20} = 260.$$

Shorcut method

1. The Average of two or more groups taken together.

(a) If the number of quantities in two groups are n_1 and n_2 and their average is x and y , respectively, the combined average (average of all of them put together) is $\frac{n_1x+n_2y}{n_1+n_2}$.

(b) If the average of n_1 quantities is x , and the average of n_2 quantities out of them is y , the average of the remaining group (rest of the quantities) is $\frac{n_1x-n_2y}{n_1-n_2}$.

2. If x is the average of $x_1, x_2, \dots, x_n$, then

- (a) The average of $x_1 + a, x_2 + a, \dots, x_n + a$ is $x + a$.
- (b) The average of $x_1 - a, x_2 - a, \dots, x_n - a$ is $x - a$.
- (c) The average of $ax_1, ax_2, \dots, ax_n$ is ax , provided $a \neq 0$.

- (d) The average of $\frac{x_1}{a}, \frac{x_2}{a}, \frac{x_3}{a}, \dots, \frac{x_n}{a}$, is $\frac{x}{a}$, provided $a \neq 0$.

Example-2: The average weight of 24 students of section A of a class is 58 Kg, whereas the average weight of 26 students of section B of the same class is 60.5 Kg. Find out average weight of all the 50 students of the class.

Solution: Here, $n_1 = 24$, $n_2 = 26$, $x = 58$ and $y = 60.5$

$$\begin{aligned}\therefore \text{The average weight of all 50 students} &= \frac{n_1x + n_2y}{n_1 + n_2} \\ &= \frac{24 \times 58 + 26 \times 60.5}{24 + 26} \\ &= \frac{1392 + 1573}{50} \\ &= \frac{2965}{50} = 59.3 \text{ Kg}\end{aligned}$$

Example-3: Average salary of all the 50 employees including 5 officers is Rs. 850. If the average salary of the officers is Rs. 2500, find the average salary of the remaining staff of the company.

Solution: Here, $n_1 = 50$, $n_2 = 5$, $x = 850$ and $y = 2500$

$$\begin{aligned}\therefore \text{The average salary of the remaining staff} &= \frac{n_1x - n_2y}{n_1 - n_2} \\ &= \frac{50 \times 8500 - 5 \times 2500}{50 - 5} \\ &= \frac{42500 - 12500}{45} \\ &= \frac{30000}{45} = 667 \text{ Kg}\end{aligned}$$

Example-4: The Average value of six numbers 7, 12, 17, 24, 26 and 28 is 19. If 8 is added to each number, what will be the new average?

Solution: The new average = $19+8=27$

Example-5: The average of x numbers is $5x$. If each number is multiplied by 8, find the average of a new set of numbers.

Solution: The average of a new set of numbers = $ax = 8 \times 21 = 168$

Exercise :

1. A grocer has a sale of Rs 6435, Rs. 6927, Rs. 6855, Rs. 7230 and Rs. 6562 for 5 consecutive months. How much sale must he have in the sixth month so that he gets an average sale of Rs, 6500 ?

- A) 4991 B) 5467
C) 5987 D) 6453

Answer: A) 4991

Explanation: Total sale for 5 months = Rs. (6435 + 6927 + 6855 + 7230 + 6562) = Rs. 34009.

Required sale = Rs. [(6500 x 6) - 34009]

= Rs. (39000 - 34009)

= Rs. 4991.

2. The average of runs of a cricket player of 10 innings was 32. How many runs must he make in his next innings so as to increase his average of runs by 4 ?

- A) 76 B) 79 C) 85 D) 87

Answer: A) 76

Explanation: Average = total runs / no. of innings = 32

So, total = Average x no. of innings = 32 x 10 = 320.

Now increase in avg = 4 runs. So, new avg = 32 + 4 = 36 runs

Total runs = new avg x new no. of innings = 36 x 11 = 396

Runs made in the 11th inning = 396 - 320 = 76

3. A pupil's marks were wrongly entered as 83 instead of 63. Due to that the average marks for the class got increased by half. The number of pupils in the class is :

- A) 45 B) 40 C) 39 D) 37

Answer: B) 40

Explanation: Let there be x pupils in the class.

Total increase in marks = $(x * 1/2) = x/2$.

$$x/2 = (83 - 63) \Rightarrow x/2 = 20 \Rightarrow x = 40.$$

4. The average weight of 8 persons increases by 2.5 kg when a new person comes in place of one of them weighing 65 kg. What might be the weight of the new person ?

- A) 70 kg B) 75 kg C) 80 kg D) 85 kg

Answer: D) 85 kg

Explanation: Total weight increased = $(8 \times 2.5) \text{ kg} = 20 \text{ kg}$.

Weight of new person = $(65 + 20) \text{ kg} = 85 \text{ kg}$.

5. The average of five consecutive odd numbers is 61. What is the difference between the highest and lowest numbers :

- A) 4 B) 8 C) 12 D) 16

Answer: B) 8

Explanation: Let the numbers be $x, x + 2, x + 4, x + 6$ and $x + 8$.

Then $[x + (x + 2) + (x + 4) + (x + 6) + (x + 8)] / 5 = 61$.

or $5x + 20 = 305$ or $x = 57$. So, required difference = $(57 + 8) - 57 = 8$

Sequence: Succession of numbers arranged in a definite order forming a definite pattern is known as sequence.

Series: If $a_1, a_2, a_3, a_4, \dots, a_n, \dots$ is a sequence, then the expression

$a_1 + a_2 + a_3 + a_4 + \dots + a_n + \dots$ is a series.

A series is finite or infinite according to as the number of terms in the corresponding sequence is finite or infinite.

2.2 Progressions:

Progressions: Those sequences whose terms follow certain patterns are called progressions.

2.2.1 Arithmetic Progression (A.P.):

Arithmetic Progression (A.P.): A sequence is called an Arithmetic Progression if the difference between two consecutive terms is always same.

i.e., $a_{n+1} - a_n = \text{constant} = d$ for all $n \in N$

The constant difference, generally denoted by 'd' is called the common difference.

a_n is called the n^{th} or last term of an A.P.

$$a_n = l = a + (n - 1)d$$

(i) Three consecutive, terms in an A.P are taken as $a - d, a, a + d$.

(ii) Four consecutive terms in an A.P taken as $a - 3d, a - d, a + d, a + 3d$.

Note : If each term of an A.P. is (increased/ decreased) by K then A.M. is also (increased/ decreased) by K.

If each term of an A.P. is (multiplied/Divided) by K, then A.M is also (multiplied/Divided) by same number K.

Rule 1. Let a be the first term and d be the common difference of an A.P. Then its n^{th} term is

$$a + (n - 1)d, \text{ i.e., } a_n = a + (n - 1)d.$$

Rule 2. The sum S_n of n terms of an A.P. with first term is ' a ' and common difference is ' d ' is

$$S_n = \frac{n}{2} [2a + (n - 1)d] \text{ or } S_n = \frac{n}{2} [a + l], \text{ where } l = \text{last term} = a + (n - 1)d.$$

2.2.2 Geometric Progression :

A sequence of non-zero numbers is called a Geometric Progression (abbreviated as G.P.) if the ratio of a term and the term preceding to it is always same. The constant ratio is called the common ratio (r) of the G.P.

In other words, a sequence $a_1, a_2, a_3, \dots, a_n$ is called a Geometric Progression if

$$\frac{a_{n+1}}{a_n} = \text{Constant } \forall n \in N$$

Three numbers in G.P is taken as a, ar, ar^2 or $\frac{a}{r}, a, ar$

Geometric Series : If $a_1, a_2, a_3, a_4, \dots, a_n, \dots$ is a G.P., then the expression $a_1 + a_2 + a_3 + a_4 + \dots + a_n + \dots$ is called a geometric series.

Rule 3. The n^{th} term of a G.P. with first term a and common ratio r is given by

$$a_n = ar^{n-1}.$$

Rule 4. The sum of n terms of a G.P. with first term ' a ' and common ratio ' r ' is given by

$$S_n = a \left(\frac{1 - r^n}{1 - r} \right) \text{ for } r < 1$$

$$S_n = a \left(\frac{r^n - 1}{r - 1} \right) \text{ for } r > 1$$

In fact these two are exactly identical. The only thing which must be noted is that the above formulas do not hold for $r = 1$, the sum of n terms of the G.P. is $S_n = na$. where $r = 1$.

Rule 5. The sum of an infinite G.P. with 1st term is ' a ' and common ratio is r ($-1 < r < 1$ i.e., $|r| < 1$) is given by

$$S_{\infty} = \frac{a}{1 - r}$$

Rule 6. Three non-zero numbers a, b, c are in G.P. if $b^2 = ac$ or $b = \sqrt{ac}$.

Here, b is known as the Geometric Mean of a and c .

2.2.3 Harmonic Progression:

Harmonic Progression: If a, b, c, d , are in H.P. then, $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}, \frac{1}{d}$ will form an A.P. and then we can apply all rules of A.P.

$$n^{th} \text{ term of HP} = \frac{1}{n^{th} \text{ term of AP}}$$

Note: No term of HP can be zero.

Some Special Sequences:

- 1) The sum of first n natural numbers,

$$\sum n = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

- 2) The sum of squares of first n natural numbers

$$\sum n^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

- 3) The sum of cubes of first n natural numbers

$$\sum n^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

Exercise:

Show that the series 1,3,5,7,9, ...in Arithmetic progression

Find the nth term and 19th term of the sequence 5,2, -1, -4, ...

Find the sum of the series .5 + .51 + .52 + ... to 100 terms.

Find the sum of 20 terms of an A. P. whose first term is 3 and last term is 57

Find the sum of 8 terms and n terms of the sequence 9, -3, 1, - $\frac{1}{3}$, ...

Find the 100th of the sequence 1, $\frac{1}{3}$, $\frac{1}{5}$, $\frac{1}{7}$, ...

Exercise

1. Determine 25th term of an A.P. whose 9th term is -6 and common difference is 5/4.

(a) 16 (b) 18 (c) 12 (d) 14

Answer: (d) 14

Explanation: $d = \frac{5}{4}$, $a_9 = -6$

$$a_n = a + (n-1)d \Rightarrow a_9 = a + 8d \Rightarrow -6 = a + 8\left(\frac{5}{4}\right)$$

$$\Rightarrow a = -16$$

The 25th term $a_{25} = a + 24d$

$$\Rightarrow a_{25} = -16 + 24\left(\frac{5}{4}\right) = 14$$

2. Which term of the A.P. 5, 13, 21, ... is 181?

- (a) 21st (b) 22nd (c) 23rd (d) 24th

Answer: (c) 23rd

Explanation: Here, first term $a = 5$

Common difference $d = 8$

Let,

181 be the n th,

i.e., $a_n = 181$

$$\therefore 181 = 5 + (n - 1)8 \text{ or, } 176 = (n - 1)8$$

$$\therefore n - 1 = 176 \div 8 = 22$$

$$\therefore n = 23$$

Hence, 181 is the 23rd term.

3. Determine k so that $2/3$, k and $5k/8$ are the three consecutive terms of an A.P.

- (a) $16/33$ (b) $14/33$ (c) $12/33$ (d) $18/33$

Answer: (a) $16/33$

Explanation: $\frac{2}{3}, k, \frac{5}{8}k$ be 3 terms of A.P.

Common difference between two consecutive terms is equal.

$$k - \frac{2}{3} = \frac{5}{8}k - k \Rightarrow 2k - \frac{5}{8}k = \frac{2}{3} \Rightarrow k = \frac{16}{33}$$

4. If the 12th term of an A.P. is -13 and the sum of the first four terms is 24, then what is the sum of the first 10 terms?

- (a) 0 (b) 2 (c) 1 (d) 4

Answer: (a) 0

Explanation: $a_n = a + (n - 1)d \Rightarrow a_{12} = a + 11d$

$$\Rightarrow -13 = a + 11d \text{ (1)}$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] \Rightarrow S_4 = \frac{4}{2}[2a + 3d]$$

$$\Rightarrow 12 = 2a + 3d \text{ (2) Solving}$$

$$\text{___(1) and ___(2)}$$

$$a = 9, d = -2$$

$$S_{10} = \frac{10}{2}[2a + 9d] = 0$$

5. How many terms are there in an A.P. whose first and fifth terms are -14 and 2, respectively and the sum of terms is 40?

- (a) 15 (b) 5 (c) 10 (d) 20

Answer: (c)

Explanation: Here, $a = -14$

Let, d be the common difference

$$a_5 = 2 \Rightarrow a + 4d = 2 \Rightarrow -14 + 4d = 2$$

$$\therefore d = 4$$

Let 40 be the sum of n terms of this A.P.

$$S_n = \frac{n}{2}[2a + (n-1)d] \Rightarrow 40 = \frac{n}{2}[2(-14) + (n-1)4]$$

$$\Rightarrow (n+2)(n-10) = 0$$

$\therefore n = 10$ or, -2 . But $n \neq -2$.

Hence, the required number of terms are 10.

6. Find the seventh term of the series $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots$

- (a) $\frac{1}{120}$ (b) $\frac{1}{128}$ (c) $\frac{1}{144}$ (d) $\frac{1}{121}$

Answer: (b)

7. How many terms are there in the G.P. 1, 2, 4, 8, ..., 4096?

- (a) 14 (b) 13 (c) 12 (d) 15

Answer: (b)

8. Find the sum of the series $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots \infty$.

- (a) 1 (b) 12 (c) 15 (d) 13

Answer: (a)

9. What term of progression 18, -12, 8, ... is 512 ? 729

- (a) 15 (b) 18 (c) 9 (d) 12

Answer: (c)

10. The 3rd term of G.P. is the square of the first term. If the 2nd term is 8, determine the 6th term.

- (a) 136 (b) 132 (c) 128 (d) 124

Answer: (c)